Differential Equations

Solving by separation of variables

1 Solve $\dfrac{dy}{dx} = \dfrac{y}{x}$ for $x > 0$, given $y = 6$ when $x = 2$.

2 Solve $\dfrac{dy}{dx} = 3x^2y$, given $y = 2$ when $x = 0$.

3 Solve $\dfrac{dy}{dx} = \dfrac{2x}{y^2}$, given $y = 2$ when $x = 0$.

4 Solve $\dfrac{dy}{dx} = \dfrac{y^2}{x}$ for $x > 0$, given $y = 1$ when $x = 1$.

Forming differential equations from a description

5 The rate of decrease of a quantity Q is proportional to Q itself. Write a differential equation for $\dfrac{dQ}{dt}$, using $k > 0$ as the constant of proportionality.

6 A tank drains so that the rate of decrease of its volume V is proportional to $\sqrt{V}$. Write a differential equation for $\dfrac{dV}{dt}$, using $k > 0$.

7 A chemical reaction's concentration C decreases at a rate proportional to the square of the concentration. Write a differential equation for $\dfrac{dC}{dt}$, using $k > 0$.

Solving modeling equations with initial conditions

8 A population P satisfies $\dfrac{dP}{dt} = 0.08P$. If $P = 500$ when $t = 0$, find P when $t = 10$, to the nearest whole number.

9 Solve $\dfrac{dV}{dt} = -k\sqrt{V}$ (from the tank problem above), given $V(0) = V_0$.

First-order linear differential equations using an integrating factor

10 Solve $\dfrac{dy}{dx} + 3y = 12$ using an integrating factor, given $y = 2$ when $x = 0$.

11 Solve $\dfrac{dy}{dx} - 2y = e^{3x}$ using an integrating factor, given $y = 1$ when $x = 0$.

12 Solve $\dfrac{dy}{dx} + y = 5$ using an integrating factor, given $y = 0$ when $x = 0$.

13 Solve $\dfrac{dy}{dx} + 4y = 8$ using an integrating factor, given $y = 3$ when $x = 0$.

Further integrating factor examples

14 Solve $\frac{dy}{dx} + \frac{y}{x} = x^2$ for $x > 0$ using an integrating factor, given $y = 1$ when $x = 1$.

15 Solve $\frac{dy}{dx} - \frac{2y}{x} = x^3$ for $x > 0$ using an integrating factor, given $y = 3$ when $x = 1$.

Verifying solutions to differential equations

16 Verify that $y = 3e^{2x}$ satisfies $\frac{dy}{dx} - 2y = 0$.

17 Verify that $y = x^2 + Ce^{-x}$ is a general solution of $\frac{dy}{dx} + y = x^2 + 2x$.

18 Verify that $y = \frac{1}{2}\sin(2x)$ satisfies $\frac{d^2y}{dx^2} + 4y = 0$.

General and particular solutions

19 Find the general solution of $\frac{dy}{dx} = 4x^3 - 6x$, and explain what determines a particular solution from this family.

20 A curve satisfies $\frac{dy}{dx} = 6x^2 - 4$ and passes through $(1, -1)$. Find the equation of this particular solution.

21 A curve satisfies $\frac{dy}{dx} = \frac{1}{x}$ for $x > 0$ and passes through $(1, 4)$. Find the particular solution.

Applications: Newton's law of cooling

22 A cup of tea at 85° C is placed in a room at 22° C. Using $T(t) = T_a + (T_0 - T_a)e^{-kt}$, and given that after 8 minutes the tea has cooled to 65° C, find k, to three decimal places.

23 Using the value of k found above, find the temperature of the tea after 20 minutes, to the nearest degree.

24 A metal rod at 200° C cools in air at 25° C, with cooling constant $k = 0.03$ per minute. Find its temperature after 15 minutes, to the nearest degree.

A well-designed word problem: a mixing tank

25 This question has three parts. A tank contains 300 litres of brine with 10 kg of dissolved salt. Fresh water flows in at 6 litres/minute, and the well-mixed solution flows out at the same rate, keeping the volume constant. Let $S(t)$ be the mass of salt (kg) at time t minutes. (a) Explain why $\frac{dS}{dt} = -\frac{S}{50}$. (b) Solve this differential equation for $S(t)$, given $S(0) = 10$. (c) Find how long it takes for the salt content to drop to 2 kg, to the nearest minute.